An ensemble of open binary CALPHAD assessments quantifies the uncertainty of the $Nd_2Fe_{14}B$ feasibility threshold and surfaces inconsistencies invisible to single-assessment users

I. Radulov

7 November 2026

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

Abstract

A single merged ternary description built from open binary CALPHAD assessments is only one of many that can be constructed from the published literature. For Nd–Fe–B, three Fe–Nd, seven Fe–B and four Nd–B assessments are freely available, giving 84 candidate ternary descriptions. We screen all 84 for feasibility of $Nd_2Fe_{14}B$ at the stoichiometric composition over 800–1600 K and report the distribution of feasibility thresholds. Seventy-eight of the 84 combinations (92.9 %) are mutually compatible under a strict (1 J mol^{-1}) merge-then-edge-check protocol. The six failures all involve the Witusiewicz 2019 Fe–B assessment—the only one that treats boron as purely interstitial in the disordered bcc, fcc and hcp phases—and arise because the merge silently alters the sublattice model of those phases; each fails by only 6.2 J mol^{-1} at a single test point. Re-admitting them leaves the range unchanged and shifts no median by more than 0.07 kJ. At 800 K the ensemble threshold spans −9.26 to −8.33 kJ/mol (per mole of atoms; median −8.79, spread 0.94). The two available full ternary assessments (Hallemans 1995 and Chen 2019) both fall inside this range, with Hallemans coinciding with the median to within 0.01 kJ. The single-value threshold of −9.2 kJ/mol used in our prior Stage 3 work sits at the more-negative extreme of the ensemble. The spread grows from 0.94 kJ at 800 K to 2.35 kJ at 1600 K. We argue that the ensemble is a minimal-assumption description of the current open thermodynamic data for Nd–Fe–B, and that single-value CALPHAD claims should be reported together with the ensemble spread around them.

1 Introduction

The CALPHAD method has produced more than fifty years of thermodynamic assessments for binary alloy systems. For well-studied binaries, several independent assessments are typically available—each fitted against the data visible to its authors at the time, each a point estimate without a formal uncertainty. In the ternary case, where the number of binary mergers grows combinatorially with the number of available binary assessments, a user who selects a single merged description implicitly selects one point from a larger population. The spread of that population is hidden.

In earlier work on Nd–Fe–B (Radulov et al., Stage 3) we reported a feasibility threshold for $Nd_2Fe_{14}B$ constructed from a single merged binary description based on one choice per edge. The reported value of −9.2 kJ/mol at 800 K was consistent with the two published full ternary assessments (Hallemans 1995 and Chen 2019) to within 0.1–0.4 kJ at 800 K. We did not then quantify

what fraction of the open-literature ensemble the single value represents, nor what the spread would be if every open combination were examined.

Here we perform that calculation. We screen all 84 binary-assessment combinations available in the SGTE binary library for the Fe–Nd–B system, apply a merge-then-edge-check protocol, and compute the $Nd_2Fe_{14}B$ feasibility threshold for every surviving combination at eight temperatures. The resulting distribution quantifies two quantities that any single merged description leaves implicit: the mutual compatibility of the open binary data for Nd–Fe–B, and the sensitivity of the missing-ternary $Nd_2Fe_{14}B$ feasibility threshold to the choice of binary assessment.

2 Method

2.1 Assessment inventory

The binary TDB files used in this work are part of the open SGTE collection of binary datasets [1] and are also distributed, together with calculated phase diagrams, by the ORCHID project [2, 3] (`https://data.orchid-project.org`). The SHA-256 digests of the four Nd–B files in Supplementary Table S1a agree with those published by ORCHID. We use them as maintained in our project repository (`https://github.com/iliya-radulov/my_db_project`; archived release `v4.1-stage4_trackA`: DOI 10.5281/zenodo.24039719). For Nd–Fe–B this collection provides:

- Fe–Nd: 3 assessments: `FeNd-16Che` (Chen 2016 [4]), `FeNd-16Wan` (Wang 2016 [5]), `FeNd-95Hal-LB` (Hallemans 1995 [6], as recorded in Landolt–Börnstein).
- Fe–B: 7 assessments: `FeB-01Pal` (Palumbo 2001 [7]), `FeB-02Rom` (Van Rompaey 2002 [8]), `FeB-08Yos` (Yoshitomi 2008 [9]), `FeB-19Wit` (Witusiewicz 2019 [10]), `FeB-20Tak` (Takahashi 2020 [11]), `FeB-91Pan` (Pan 1991, unpublished [12]), `FeB-94Hal-LB` (Hallemans 1994 [13], Landolt–Börnstein).
- Nd–B: 4 assessments: `NdB-16Zho-corrected` (Zhou 2016 [14], as corrected), `NdB-19Che` (Chen 2019 [15]), `NdB-95Hal-LB` (Hallemans 1995 [6], Landolt–Börnstein), `NdB-98Che` (Chevalier 1998, unpublished [16]).

The Cartesian product yields 84 candidate merged descriptions. The complete list of binary files and their SHA-256 digests is given in Supplementary Table S1a so that the ensemble is exactly reproducible.

Naming convention. Throughout, binary assessments are referred to by their file identifiers (edge, two-digit year, first letters of the first author: `FeB-19Wit`, `NdB-19Che`, ...); author–year labels appear only in the list above. The two full ternary assessments are written *Hallemans 1995 (ternary)* and *Chen 2019 (ternary)* to distinguish them from the binary files of the same authors.

2.2 Merge-then-edge-check protocol

For each of the 84 combinations we:

1. Merge the three thermodynamic database (TDB) files into one ternary description using

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

a parsed-database merge (not text concatenation). Function and parameter conflicts are reported; shared phases are united constituent-wise.

2. Verify that the merged description reproduces each binary file along its edge to within 1 J mol^{-1} on the Gibbs energy of equilibrium. This tolerance is deliberately tight relative to typical CALPHAD parameter precision and to the numerical noise of the equilibrium solver, so that any detectable change in a binary edge is flagged as an inconsistency. Combinations that fail this edge check at any temperature are recorded but excluded from the subsequent feasibility screen.

3. For the surviving combinations, compute the $Nd_2Fe_{14}B$ feasibility threshold at eight temperatures (800, 1000, 1100, 1200, 1300, 1375, 1450, 1600 K). These span the practical processing window of Nd–Fe–B magnets (sintering, annealing and high-temperature liquid-phase regimes) while remaining computationally tractable.

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

The *feasibility threshold* $\Delta G_f^*(T)$ is the Gibbs energy of formation, relative to the SER (stable element reference) state, that a stoichiometric line compound at the $Nd_2Fe_{14}B$ composition ($x_{Fe} = 14/17$, $x_{Nd} = 2/17$, $x_B = 1/17$) must reach to become competitive with the equilibrium phase assemblage of the merged binary description:

$$\Delta G_f^*(T) = \frac{1}{1000}\Big(G_{\text{competing}}(T) - \sum_i x_i\, G_i^{\text{SER}}(T)\Big) \tag{1}$$

in kJ/mol (per mole of atoms; this unit is used throughout), where $G_{\text{competing}}$ is the equilibrium Gibbs energy per mole of atoms (J mol^{-1}) of the merged binary description at that composition and $G_i^{\text{SER}}(T)$ are the SGTE `GHSER` reference functions of Fe, Nd and B. The compound is favoured when its own formation Gibbs energy satisfies $\Delta G_f \le \Delta G_f^*$. A more negative threshold therefore means that the binary phases are already strongly stable and the compound must be strongly stabilised (by ternary interactions or by its own Gibbs-energy function) to appear; a less negative threshold means that weaker stabilisation suffices. The quantity is a Gibbs-energy difference evaluated at temperature T; it was denoted ΔH_f^* in Stage 3.

The screen uses the open-source Python CALPHAD library pycalphad [17] (version 0.11.2) and our own `tdb_merge.py` utility. The threshold kernel employs the GHSER-based reference identified as the correct one in Stage 3 §6.2.

2.3 Reproducibility

Code is available under `stage4_step1/run_batch_v2.py` (dependency `stage_three/tdb_merge.py` with the `tag` argument). Results are written per combination to `results_v2/batch_results.csv` and `results_v2/detail/<combo_id>.json`. A summary is written to `results_v2/summary.json`. The 84 combinations were executed with 8 parallel workers. The per-combination run times sum to 7.1 min (5.1 s on average, 7.9 s at most), and the first and last combinations started 51 s apart, so the full screen took about one minute of wall-clock time on the hardware used.

3 Results

3.1 Survival

Of the 84 candidate combinations, 78 (92.9 %) survive the merge-then-edge-check protocol. The six failures are all combinations of FeB-19Wit with NdB-19Che or NdB-16Zho-corrected (three Fe–Nd choices × two Nd–B choices). The failure mechanism is analysed in §3.4. Re-admitting the six at a relaxed tolerance (their worst deviation is 6.2 J mol^{-1}) leaves the minimum and maximum of the threshold distribution unchanged at all eight temperatures and shifts the median by at most 0.07 kJ (for example $-8.76 \rightarrow -8.69$ at 1600 K); the six lie between the 13th and 97th percentile of the strict ensemble. The conclusions therefore do not depend on the edge-check tolerance.

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

3.2 Distribution of the feasibility threshold

Table 1 summarises the distribution of the $Nd_2Fe_{14}B$ feasibility threshold ΔG_f^* over the 78 surviving combinations at the eight temperatures.

Table 1: Distribution of the feasibility threshold ΔG_f^* (kJ/mol, per mole of atoms) over the 78 survivors; spread = max − min.

T (K)	min	q25	median	q75	max	spread
800	−9.26	−8.87	−8.79	−8.70	−8.33	0.94
1000	−8.28	−8.03	−7.88	−7.76	−7.47	0.81
1100	−8.00	−7.67	−7.49	−7.27	−7.04	0.96
1200	−7.81	−7.36	−7.17	−6.76	−6.58	1.22
1300	−7.77	−7.17	−6.94	−6.47	−6.07	1.70
1375	−8.02	−7.30	−7.06	−6.53	−5.99	2.03
1450	−8.55	−7.80	−7.45	−7.00	−6.33	2.22
1600	−9.87	−9.34	−8.76	−8.38	−7.51	2.35

Quartiles use linear interpolation between the 78 ordered values; per-combination values at all eight temperatures are provided in the supplementary data files.

The spread grows by a factor of approximately 2.5 between 800 K and 1600 K, reflecting the loss of calorimetric constraint at high temperatures and the accumulated extrapolation uncertainty of the binary assessments. To see which edge drives the spread we fitted, at each temperature, an additive model with the three edge choices as categorical factors and report each factor's sum of squares as a fraction of the total variance; the remainder is the interaction term (4 % at 800 K, 7 % at 1000 K, at most 1 % from 1300 K). The Fe–Nd choice explains the largest share at 800 K (51 %, against 26 % for Fe–B and 20 % for Nd–B). At 1000 K, the temperature of Figs. 1 and 3, the shares are 30, 36 and 26 % (Fe–Nd, Fe–B, Nd–B), and the Fe–B choice dominates from 1300 K upward (67, 58, 56 and 66 % at 1300, 1375, 1450 and 1600 K).

3.3 Comparison with full ternary assessments

We applied the same feasibility screen to the two full ternary CALPHAD assessments available in the open literature for Nd–Fe–B: Hallemans 1995 (ternary [6]; `bfend_hal_en.tdb`) and Chen 2019

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

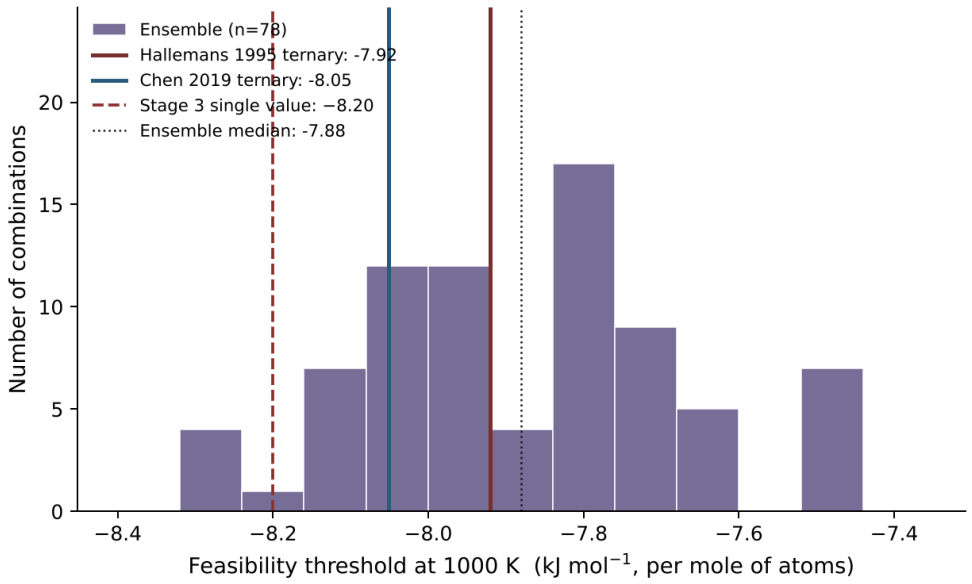


Figure 1: Histogram of ΔG_f^* at 1000 K over the 78 survivors. Vertical lines mark the two available full ternary assessments (Hallemans 1995 and Chen 2019); the Stage 3 single value (−8.20 at 1000 K) is shown as a dashed line for comparison.

(ternary [15]; `Nd-Fe-B.TDB`). We also screened the NIMS Nd-mag-STAM multicomponent database as a cross-check; its reported $Nd_2Fe_{14}B$ threshold is identical to Hallemans 1995 to 10^{-4} kJ at every temperature. This confirms that the STAM database inherits its Fe–Nd–B subsystem from Hallemans 1995 and does not constitute a third independent ternary assessment. We therefore report Hallemans 1995 and Chen 2019 as the two independent references. For both assessments the native $Nd_2Fe_{14}B$ phase was suppressed so that the computed threshold is directly comparable to the binary-ensemble threshold.

Table 2: Full ternary assessments placed on the ensemble distribution (kJ/mol, per mole of atoms).

T (K)	Hallemans 1995	Chen 2019	Ensemble median	Ensemble range
800	−8.78	−9.10	−8.79	−9.26 . . . − 8.33
1000	−7.92	−8.05	−7.88	−8.28 . . . − 7.47
1100	−7.51	−7.53	−7.49	−8.00 . . . − 7.04
1200	−7.12	−7.04	−7.17	−7.81 . . . − 6.58
1300	−6.75	−6.56	−6.94	−7.77 . . . − 6.07
1375	−6.73	−6.41	−7.06	−8.02 . . . − 5.99
1450	−7.20	−6.75	−7.45	−8.55 . . . − 6.33
1600	−8.57	−8.01	−8.76	−9.87 . . . − 7.51

Three features of the comparison are worth emphasising.

1. **At low temperature, both ternary assessments sit inside the ensemble bulk.** At 800 K and 1000 K the Hallemans 1995 ternary coincides with the ensemble median to within 0.01 and 0.04 kJ/mol respectively. The Chen 2019 ternary sits approximately one histogram-bin to the more-negative side of the median (−9.10 versus −8.79 at 800 K). Both references

fall well inside the ensemble range. The ensemble is therefore internally calibrated against an independent full ternary assessment; the median is not merely a statistical convenience but coincides quantitatively with what a careful ternary fit of the same open data produces.

2. **The Stage 3 single-point value sits at the more-negative extreme of the ensemble.** The Stage 3 paper reported −9.20 kJ/mol at 800 K from a single merged binary description. In the present ensemble this value lies in the most-negative histogram bin of the distribution—between the Chen 2019 ternary reference (−9.10) and the ensemble minimum (−9.26)—and 0.41 kJ below the ensemble median (−8.79). A typical open-assessment combination therefore yields a threshold approximately 0.4 kJ less negative than the value previously reported. The Stage 3 result was an honest consequence of the specific binary choice made at the time, but it was not a typical draw from the open literature. Future readers of the Stage 3 paper should carry this context.

3. **At high temperature the ensemble is more pessimistic than both ternaries.** From 1200 K upward both ternary thresholds lie above (are less negative than) the ensemble median at every temperature computed: Hallemans 1995 by 0.05–0.33 kJ and Chen 2019 by 0.13–0.75 kJ between 1200 and 1600 K (Table 2; for example −6.73 versus −7.06 at 1375 K and −8.57 versus −8.76 at 1600 K). At 800–1100 K, in contrast, Hallemans lies within 0.04 kJ of the median and Chen at or below it, so the crossover lies between 1100 and 1200 K, where both ternaries sit within 0.13 kJ of the median. The offset grows with temperature and is largest for Chen 2019, which lies at the 81st–87th percentile of the ensemble across 1375–1600 K, reaching the 87th at 1600 K.

   Since a less negative threshold means weaker stabilisation suffices, the binary ensemble is more pessimistic about $Nd_2Fe_{14}B$ stability at high temperature than the full ternary assessments. We interpret this as a modelling asymmetry rather than a disagreement to be reconciled: the full ternary assessments carry a $Nd_2Fe_{14}B$-specific Gibbs-energy function that smooths the high-temperature behaviour of the compound explicitly, whereas the binary ensemble treats $Nd_2Fe_{14}B$ as a hypothetical line compound and is therefore more sensitive to the high-temperature extrapolation of the binary assessments. Both framings are internally consistent. The ensemble's picture of uncertainty is itself temperature-dependent—its spread grows by a factor of 2.5 between 800 K and 1600 K—and it appropriately flags a single-value claim at high temperature as carrying more hidden uncertainty than one at low temperature.

3.4 Failure mechanism

The six failures are all combinations of `FeB-19Wit` (Witusiewicz 2019) with either `NdB-19Che` (Chen 2019) or `NdB-16Zho-corrected` (Zhou 2016, corrected). `FeB-19Wit` differs from the other Fe–B assessments in two ways; only the second causes failures (Fig. 4). The full diagnostic chain, including the two hypotheses that were tested and rejected, is documented in Supplementary Information S2.

Structural difference — an additional Fe–B polymorph. `FeB-19Wit` models the Fe–B mono-boride as two crystallographic polymorphs, `MB_B27` (FeB-type, *Pnma*) and `MB_B33` (CrB-type, *Cmcm*); all six other Fe–B assessments model only `MB_B27`. The extra phase is carried into every merged description containing `FeB-19Wit`, but it does not by itself cause an edge-check failure: all six combinations of `FeB-19Wit` with `NdB-95Hal-LB` or `NdB-98Che` pass, and the merged descriptions of those six reproduce the `FeB-19Wit` edge within tolerance. The extra polymorph therefore makes the `FeB-19Wit` ensemble members structurally different, but it is not the source of the failures.

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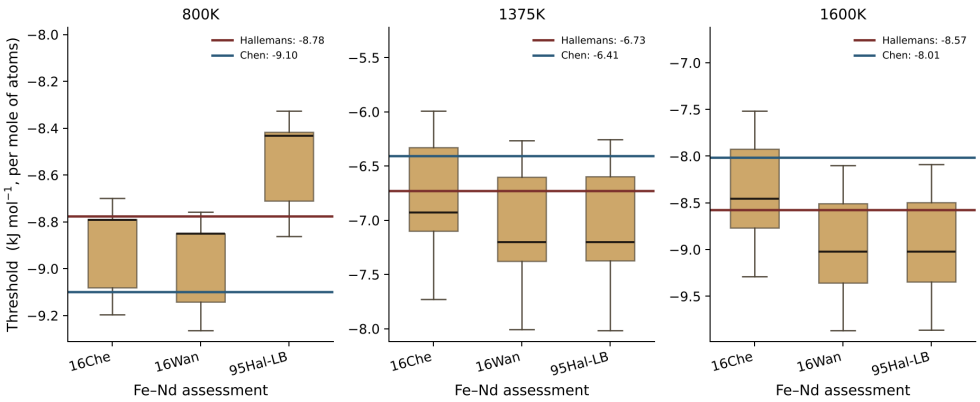


Figure 2: Box plot of the ensemble grouped by Fe–Nd assessment at three temperatures (800, 1375 and 1600 K). Hallemans 1995 (ternary) and Chen 2019 (ternary) references are overlaid as horizontal lines.

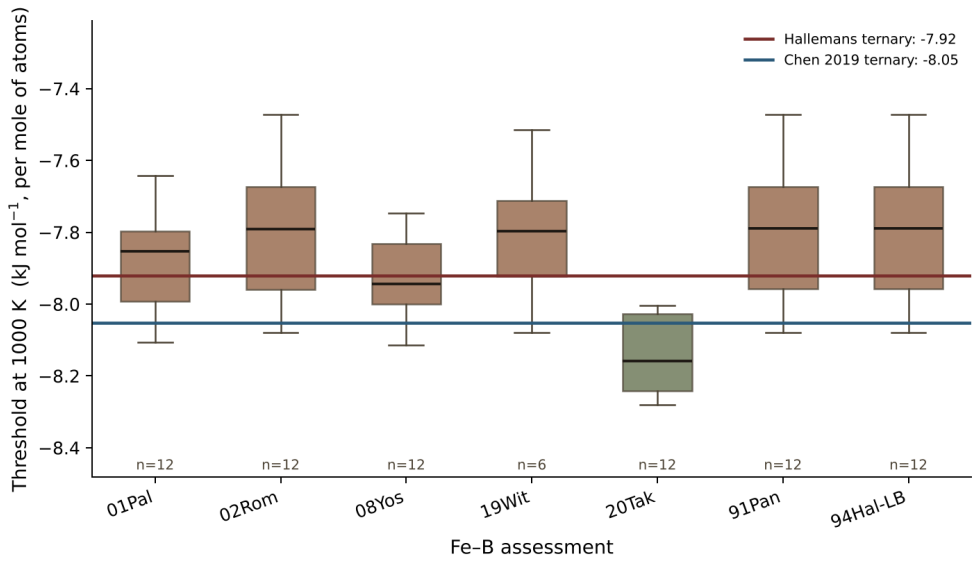


Figure 3: Box plot of the ensemble grouped by Fe–B assessment at 1000 K. Takahashi 2020 produces the tightest cluster (spread 0.28 kJ) and the most negative median (−8.16); the other six Fe–B assessments cluster near −7.80 with spreads of 0.37–0.61. Takahashi's median is more negative than either ternary reference at this temperature, by 0.11 kJ relative to Chen 2019 (−8.05) and 0.24 kJ relative to Hallemans 1995 (−7.92). The ranking illustrates which binary edge carries the largest systematic shift on the ternary feasibility threshold.

Failure mechanism — silent change of the phase model. The binary assessments use two incompatible sublattice models for boron in the shared phases FCC_A1, BCC_A2 and HCP_A3 (Table 3): a *substitutional* model, $(\mathrm{B}, M)(\mathrm{VA})$ with M = Fe or Nd, and an *interstitial* model, $(M)(\mathrm{B}, \mathrm{VA})$. `FeB-19Wit` is the only Fe–B file using the interstitial form; the other six use the substitutional form. Among the Nd–B files, `NdB-19Che` and `NdB-16Zho-corrected` use the substitutional form and `NdB-95Hal-LB` and `NdB-98Che` the interstitial form. The parsed-database merge unites the constituents of shared phases. Merging `FeB-19Wit` with a substitutional Nd–B file therefore turns, for example, FCC_A1 from $(\mathrm{Fe})(\mathrm{B}, \mathrm{VA})$ into $(\mathrm{B}, \mathrm{Fe}, \mathrm{Nd})(\mathrm{B}, \mathrm{VA})$: boron gains a position on the first sublattice, together with the end-member parameters supplied by the Nd–B file (e.g. `G(FCC_A1,B:VA;0)` $= 43514 + \mathrm{GHSERBB} - 12.217\,T$ $\mathrm{J\,mol^{-1}}$), which `FeB-19Wit` alone does not contain. The merge adds four such parameters on the Fe–B edge, redefines no symbol of `FeB-19Wit`, and reports no conflict for them. The Fe–Nd and Nd–B edges are reproduced to ≤ 0.13 $\mathrm{J\,mol^{-1}}$, but the Fe–B edge deviates by 6.21 $\mathrm{J\,mol^{-1}}$ at one of 30 test points (1600 K, $x_{\mathrm{Fe}} = 0.95$, FCC_A1 + liquid in both descriptions), with no phase-set mismatch; all six failing combinations show the same deviation at the same point. Restoring the constituents of `FeB-19Wit` on the three merged phases removes the deviation completely (0.0 $\mathrm{J\,mol^{-1}}$ at all 30 points, with or without the four added parameters), whereas deleting only the four added parameters leaves 2.5 $\mathrm{J\,mol^{-1}}$. The unified constituent set is therefore the cause, and the added end-member parameters are part of its effect. The change is silent because the merge reports it neither as a conflict nor as an error; it is visible only to an edge check against the original binary.

The same union produces a mixed model in 36 of the 78 survivors (a substitutional Fe–B file with an interstitial Nd–B file); these pass the edge check, so their induced energy change is below the tolerance, but they are not validated beyond it.

Table 3: Boron sublattice model of the four Nd–B files and outcome of the merge with `FeB-19Wit`, whose own model is interstitial, $(\mathrm{Fe})(\mathrm{B}, \mathrm{VA})$.

Nd–B file	Boron model in FCC_A1, BCC_A2, HCP_A3	Outcome with `FeB-19Wit`
NdB-19Che	substitutional $(\mathrm{B}, \mathrm{Nd})(\mathrm{VA})$	fails
NdB-16Zho-corrected	substitutional $(\mathrm{B}, \mathrm{Nd})(\mathrm{VA})$	fails
NdB-95Hal-LB	interstitial $(\mathrm{Nd})(\mathrm{B}, \mathrm{VA})$	succeeds
NdB-98Che	interstitial $(\mathrm{Nd})(\mathrm{B}, \mathrm{VA})$	succeeds

Neither feature is a bug in pycalphad or in the published assessments considered individually. Each assessment is internally self-consistent. The failure arises from combining two assessments that model boron on different sublattices, and it would be invisible to any single-assessment user; it is surfaced only by the ensemble protocol.

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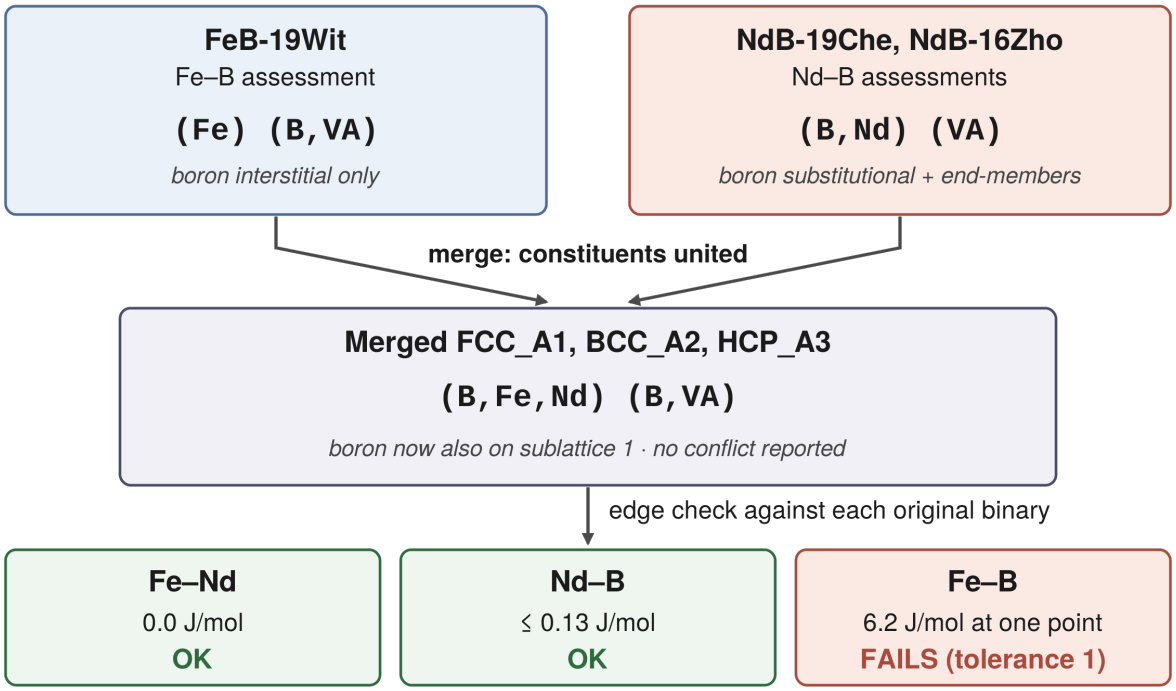


Figure 4: Schematic of the failure mechanism. `FeB-19Wit` models boron in the bcc, fcc and hcp phases as purely interstitial, (Fe)(B, VA); `NdB-19Che` and `NdB-16Zho-corrected` model it as substitutional, (B, Nd)(VA). The merge unites the constituents, giving (B, Fe, Nd)(B, VA) together with the Nd–B end-member parameters, without reporting a conflict. The Fe–Nd and Nd–B edges are reproduced, but the Fe–B edge deviates by 6.2 J mol^{-1} at one test point, exceeding the 1 J mol^{-1} tolerance. The two Nd–B files with an interstitial model (`NdB-95Hal-LB`, `NdB-98Che`) leave the Fe–B model unchanged. The additional `MB_B33` polymorph of `FeB-19Wit` does not contribute (text).

4 Discussion

4.1 The Stage 3 single value, reinterpreted

In Stage 3 we reported a feasibility threshold of −9.2 kJ/mol at 800 K from a single merged binary description. The present ensemble places that value at the more-negative extreme of the 78-member distribution, just inside the range (−9.26 to −8.33) and 0.4 kJ below the median (−8.79). The Stage 3 value was therefore an honest result of the specific binary choice made at the time, but it was not a typical draw from the open literature. A typical open-assessment combination gives a threshold about 0.4 kJ less negative—closer to the Hallemans 1995 (ternary) than to the Chen 2019 (ternary). Future readers of the Stage 3 paper should carry this context.

4.2 Hallemans at the median

The quantitative result most useful for calibration is the coincidence of Hallemans 1995 with the ensemble median. At 800 K the two agree to within 0.01 kJ. We do not claim that this reflects a particular virtue of Hallemans's assessment; the ensemble median is simply what the open binary data say on average, and a full ternary assessment that fits the same data should agree with it. The coincidence is strong evidence that the ensemble is internally calibrated against an independent full ternary assessment, not merely a parameterisation exercise.

4.3 High-temperature pessimism

From 1200 K upward both ternary assessments lie above (are less negative than) the ensemble median, and at 1375 to 1600 K Chen 2019 (ternary) sits at the 81st–87th percentile of the ensemble, reaching the 87th at 1600 K. Because a less negative threshold means weaker stabilisation suffices, the ensemble is more pessimistic about $Nd_2Fe_{14}B$ stability at high temperature. This is not a disagreement to be reconciled—both framings are internally consistent—but it is a useful reminder that the ensemble's picture of uncertainty is itself temperature-dependent. The spread grows by a factor of 2.5 between 800 K and 1600 K; a single-value claim at high temperature carries more hidden uncertainty than a single-value claim at low temperature.

4.4 Failures as a result

The six failures are a result, not an artefact, although each is a small energy mismatch (for the combination examined in detail, 6 $J\,mol^{-1}$ at a single test point) rather than a gross inconsistency. They identify `FeB-19Wit` as the only Fe–B assessment that models boron as purely interstitial in the bcc, fcc and hcp phases, and `NdB-19Che` and `NdB-16Zho-corrected` as the Nd–B members that model it as substitutional. The ensemble method surfaces these authoring inconsistencies between published open assessments; a single-user workflow would never see them. This is a modest but useful methodological byproduct of the approach.

4.5 Limitations

- The ensemble is binary-only. Full ternary interactions are not modelled.

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

- The composition is pinned at exact $Nd_2Fe_{14}B$. The screen is a line-compound stability check against the merged binary assemblage; it is not a full ternary phase-diagram calculation.
- The merge unites the constituents of shared phases, which silently changes the sublattice model whenever the two files disagree (36 survivors combine a substitutional Fe–B with an interstitial Nd–B file); the edge check bounds the effect on the binary edges but not on the ternary threshold.
- The 1 J mol^{-1} edge-check tolerance is strict; the six excluded combinations deviate by at most 6.2 J mol^{-1}. Re-admitting them does not change the conclusions (§3.1), but the headline survival rate of 92.9 % depends on this tolerance.
- The 78 survivors are treated as an unweighted population. A weighting by data quality, year, or by a posterior statistical treatment (as outlined in our ongoing methodological work) would sharpen the distribution.
- The library of open Nd–Fe–B assessments is modest ($3 \times 7 \times 4 = 84$ combinations, 78 surviving). For other alloy systems the available combinations may be larger or smaller.

zenodo:24039883 07 Nov 2026 DOI: 10.5281/zenodo.24039883

5 Code and data availability

The batch script, merge utility and threshold kernel are available at `github.com/iliya-radulov/my_db_project` under `stage_three/stage4_step1/`. The SGTE binary assessment library is referenced in Supplementary Table S1a, and the ternary files used for the overlay in Table S1b, with SHA-256 digests. The diagnostic scripts of Supplementary Information S2 are in the subfolder `diagnostics/`. The per-combination output files (CSV, JSON details and merged TDBs) are included in the archived release (release `v4.1-stage4_trackA`, DOI 10.5281/zenodo.24039719; all versions of the repository: 10.5281/zenodo.23155920).

Implementation note. A shared intermediate-filename race condition in `tdb_merge.py` under parallel execution was identified and fixed by introducing a `tag` argument, so that each of the 84 combinations writes to its own named file. Generally, parallel workers must never share a fixed intermediate filename.

6 Conclusions

Treating the open binary CALPHAD assessments of Nd–Fe–B as a population rather than selecting one combination yields a feasibility-threshold distribution for $Nd_2Fe_{14}B$ that is 0.94 kJ wide at 800 K and 2.35 kJ wide at 1600 K, and that contains both available full ternary assessments, with Hallemans 1995 (ternary) at the median at low temperature. The single-value threshold used in our prior work sits at the more-negative extreme of this distribution. The six-combination failure rate (7 %) is a tolerance-level result: it arises only when the single Fe–B assessment with an interstitial boron model is merged with the two Nd–B assessments with a substitutional one, and including the six combinations changes the distribution negligibly. The ensemble is a minimal-assumption description of the current open thermodynamic data for Nd–Fe–B.

Acknowledgements

I thank Hongbin Zhang for a fruitful discussion.

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